TUNDUMA TOWN COUNCIL

SUMAGHA SECONDARY SCHOOL

FORM TWO ASSESSMENT SERIES III

037 COMPUTER SCIENCE -- MARKING SCHEME

Year: 2026

SECTION A (15 Marks)

1. Multiple Choice (10 Marks, 1 mark each)

Q	Answer
(i)	D - By preventing unauthorized access
(ii)	A - They instantly produce images
(iii)	B - Backspace
(iv)	C - Central Processing Unit
(v)	D - Generations of computers
(vi)	C - They can be used everywhere
(vii)	A - It helps in solving complex problems efficiently
(viii)	B - A cup that can hold different liquids
(ix)	C - Main function
(x)	C - time.sleep(3)

2. Matching (05 Marks)

Item	Answer
(i)	D - World Wide Web
(ii)	E - Web page
(iii)	G - Website
(iv)	A - Search engine
(v)	F - Web browser

SECTION B (70 Marks)

3(a) Three reasons OS must load first (03 Marks)

- It manages hardware resources needed by application software
- It provides the interface through which applications communicate with hardware
- It allocates memory and CPU time to applications before they can run

3(b) Three key OS functions (03 Marks)

- Memory management - allocates and manages the computer's RAM
- Process management - controls the execution of running programs
- File management - organises and controls access to files and folders

3(c) Four types of Operating System (04 Marks)

- Batch OS - processes jobs in batches with no direct user interaction
- Time-sharing OS - allows multiple users to share system resources simultaneously
- Real-time OS - responds to input within a strict, predictable time limit
- Distributed OS - manages a group of independent computers so they act as a single system

4(a) Three rules for Python identifiers (03 Marks)

- Must start with a letter or an underscore, not a digit
- Can contain only letters, digits and underscores
- Cannot be a Python reserved keyword

4(b) Grading program (03 Marks)

```
score = int(input("Enter score: "))
if score < 0:
print("Invalid score")
elif score >= 75:
print("Grade: A")
elif score >= 65:
print("Grade: B")
elif score >= 45:
print("Grade: C")
elif score >= 35:
print("Grade: D")
else:
print("Grade: F")
```

4(c) List of cities program (03 Marks)

```
cities = ["Dodoma", "Arusha", "Mwanza"]
new_city = input("Enter a new city name: ")
cities[1] = new_city
print(cities)
```

5(a) Meaning of debugging (02 Marks)

Marking guide: Debugging is the process of finding and fixing errors (bugs) in a computer program so that it runs correctly.

5(b)(i) Three syntax errors in Juma's code (06 Marks)

- "int main(" is missing a closing parenthesis and opening brace - should be: int main() {
- "scanf("%d", num1);" is missing the address-of operator - should be: scanf("%d", &num1);
- "printf("The sum is: %d", &sum);" wrongly uses & in printf - should be: printf("The sum is: %d", sum);

5(b)(ii) Type of error when entering 10.5 (02 Marks)

Marking guide: This is a logical/runtime error caused by a type mismatch: scanf with %d expects a whole number, but 10.5 is a decimal (float) value, so it cannot be read correctly into an int variable, causing the program to behave incorrectly or crash.

6(a) Five internet connectivity methods (05 Marks)

- Dial-up
- DSL (Digital Subscriber Line)
- Wi-Fi
- Fiber optic
- Mobile/cellular data (3G/4G/5G)

6(b)(i) Method of learning (01 Mark)

Marking guide: E-learning / Distance learning (online learning).

6(b)(ii) Two advantages (02 Marks)

- Flexible learning time and location
- Access to a wide range of learning materials at any time

6(b)(iii) Two disadvantages (02 Marks)

- Lack of face-to-face interaction with teachers/peers
- Requires reliable internet access and a suitable device

7(a) Algorithm vs Pseudocode (02 Marks)

Marking guide: An algorithm is a step-by-step logical procedure for solving a problem, which can be described in plain language or diagrams. Pseudocode is a structured way of writing an algorithm using code-like statements, without following the strict syntax of any particular programming language.

7(b) Simple interest - algorithm and pseudocode (04 Marks)

Marking guide: Algorithm: 1. Start 2. Input P, R, T 3. Calculate I = (P*R*T)/100 4. Display I 5. Stop

```
START
INPUT P, R, T
I = (P * R * T) / 100
PRINT I
STOP
```

7(c) Flowchart symbols shown (04 Marks)

Marking guide: Based on the TRUE/NO labels shown on the answer sheet, the two figures are: (i) Diamond/Rhombus - Decision symbol, used to show a decision point with a Yes/No (True/False) outcome. (ii) Parallelogram - Input/Output symbol, used to show data being entered into or displayed out of the program. (Match against the actual symbols printed on the student's paper.)

8(a) Four causes of cybercrime (04 Marks)

- Weak passwords and poor security practices
- Lack of awareness/education about online safety
- Financial gain motives
- Inadequate laws and weak enforcement

8(b)(i) Meaning of netiquette (01 Mark)

Marking guide: Netiquette is a set of rules for acceptable, respectful and polite behaviour when communicating online.

8(b)(ii) Five tips for good netiquette (05 Marks)

- Be respectful and polite to others
- Avoid typing in all capital letters (it reads as shouting)
- Stay on topic in discussions/forums
- Do not spam or forward chain messages without permission
- Respect other people's privacy and personal information

9(a) Components of first four computer generations (04 Marks)

- Vacuum tubes (1st generation)
- Transistors (2nd generation)

- Integrated circuits/ICs (3rd generation)
- Microprocessors (4th generation)

9(b)(i) Importance of antistatic wrist strap (02 Marks)

Marking guide: It prevents electrostatic discharge (ESD) from the technician's body reaching and damaging sensitive electronic components during maintenance.

9(b)(ii) Four reasons for electrical troubleshooting (04 Marks)

- To identify faulty components causing system malfunction
- To prevent electrical hazards such as fire or shock
- To ensure the system functions properly and reliably
- To extend the lifespan of computer components

SECTION C (15 Marks)

10. Power fluctuation: causes, effects and protective devices (15 Marks)

- Causes: lightning strikes, overloading of electrical circuits, faulty wiring, heavy machinery/motors switching on and off, instability from the utility power grid
- Effects: computers restart unexpectedly, data loss or file corruption, hardware damage, reduced lifespan of components
- Protective devices: Uninterruptible Power Supply (UPS), surge protector, voltage regulator/stabilizer, standby generator as backup power